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Adaptive Feature Fusion and Enhancement Network for Food Nutrition Estimation

Donglin Zhang, Boyuan Ma, and Xiao-Jun Wu

***Abstract*—Nutrition estimation plays an important role in promoting food analysis and quality development. In recent years, with the development of deep learning and computer vision technologies, several vision-based nutrition evaluation methods have emerged. Compared with the traditional physical and chemical methods, these methods can quickly predict the content information of nutrients in food without damage. However, despite their promising evaluation performance, there are still some issues that need to be further conquered: first, how to effectively obtain comprehensive feature information about evaluated food, second, how to fully utilize different scale features and effectively fuse. To solve these problems, we propose an adaptive feature fusion and enhancement method, named ADFE. Specifically, the developed ADFE leverages the swin-transformer as the backbone network, which can obtain more effective features through window self-attention and shifted window self-attention. Besides, we design an adaptive fusion multimodal feature module, which can well handle the relationships between RGB and depth information. Furthermore, We propose a semantic-information enhancement module to enhance feature extraction and effectively fuse multiscale features. Finally, we conduct comprehensive experiments on the benchmark dataset Nutrition5k, compared with state-of-the-art methods, the mean value of the percentage mean absolute error (PMAE) for our method is 17.8%. The PMAE of calories, carbs, and protein is 14.4%, 21.0%, and 19.4%, marking improvements of 0.6%, 1.4%, and 1.6%. Code and models are available online https://github.com/123clown/ADFE.**

***Index Terms*—Feature fusion, food analysis, multimodal data, nutrition estimation.**

I. Introduction

OVER the past decade, some works [1], [2], [3], [4], [5] are proposed to explore the relationships between dietary patterns and disease. For example, the GBD 2017 Diet Collaborators study [1] finds that poor diet is the leading risk factor for death worldwide. Hospitalized patients automatic system [2] shows that there are some correlations between dietary patterns and depressive symptoms in middle-aged people. The above-mentioned methods indicate that poor eating patterns can lead to the occurrence of physical and mental illness, emphasizing the need for effective methods to promote healthier dietary behaviors. Nutrition estimation addresses this need by accurately predicting the nutrient content of foods (such as calories, fat, and protein) through physicochemical or deep learning methods. Therefore, estimating the nutrition of different foods plays an important role in improving poor eating patterns and guiding people to make scientific food choices. Moreover, accurate assessment of food nutrition can accelerate the development of food composition analysis and precision nutrition [6]. Given the above, the evaluation of food nutrition has received extensive attention in many research fields, such as food chemistry [7], artificial intelligence [8], [9], [10], and health management [11], [12].

Early methods of food nutrition estimation are mainly based on chromatographic and chemical methods. Specifically, these methods include gas chromatography [13], liquid chromatography [14], mass spectrometry [15], titration [16], spectrophotometry [7], etc. Generally speaking, these methods often yield good results for qualitative and quantitative analysis of certain types or food components, making a significant contribution to the research of food composition analysis. However, due to the need for additional experimental equipment, professional knowledge, and the destruction of the original structure of food, these methods are mostly used in biochemical research and industrial production, and cannot meet the needs of daily food nutrition evaluation.

Recently, with the development of computer vision [17], [18], some vision-based food recognition [19], [20] and nutrition estimation methods [8], [21], [22] are proposed. These methods can obtain nutritional information (e.g., calorie, protein quality, and fat quality) by using food images. Due to the popularity of mobile devices, vision-based methods can be well deployed and nutritional information can be quickly obtained. Then, people can eat healthily under the guidance of nutritional information. Therefore, some vision-based nutrition methods [8], [21], [22] are developed. Representative methods include, but are not limited to, swin-nutrition [21], Google-nutrition [8], and nutrition fusion network [22]. To be specific, swin-nutrition directly leverages RGB images to evaluate food nutrition, which can achieve a good performance. However, swin-nutrition does not take into account some necessary three-dimensional spatial information. To solve the problem, Google-nutrition is proposed, which not only utilizes RGB images but also depth information. Thus, spatial information can be utilized, resulting in the better performance. Afterwards, a nutrition fusion network is

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The authors are with the School of Artificial Intelligence and Computer Science, Jiangnan University, Wuxi 214122, China (e-mail: dlinzzhang@gmail.com; 6233111045@stu.jiangnan.edu.cn; wu_xiaojun@jiangnan.edu.cn).

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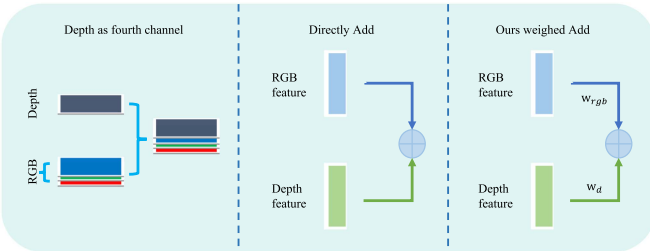

Fig. 1. Comparison of RGB-D fusion methods. The left method takes the depth image as the fourth channel and directly adds the RGB feature and the depth feature. Our method accounts for the differences between RGB features and depth features, using weighted addition scheme to fuse the RGB features and depth features.

developed, which directly adds RGB features and depth features. Moreover, Feature Pyramid Networks (FPN) [23], Bidirectional Feature Pyramid Networks (BFP) [24], and Convolutional Block Attention Module (CBAM) [25] are combined to carry out the multiscale fusion of features. The abovementioned methods confirm the effectiveness of vision-based nutritional assessment and can achieve good results. However, there are still some problems that need to be further addressed, as follows.

1) Most existing methods leverage the depth image as the fourth channel or simply add RGB features and depth features, without considering the relationships between RGB features and depth features, which may result in the loss of necessary information.
2) Most existing vision-based nutritional assessment methods usually adopt convolutional neural networks (such as ResNet [26], InceptionV2 [27]) as feature extractors. However, the inherent complexity of food images may make it difficult for CNN to extract effective information.
3) Features at different scales are not fully utilized. Existing methods usually leverage FPN to fuse multiscale features, which is a top-to-bottom fusion process that allows shallow features to contain deep semantic information. However, the abovementioned scheme can effectively extract deep semantic features, local details of food images are not well explored, resulting in suboptimal nutrition evaluation performance.

To mitigate the abovementioned problems, we develop a novel adaptive feature fusion and feature enhancement method for food nutrition estimation. Specifically, the proposed method adopts swin-transformer [28] as a feature extraction network, which can obtain better nutrition-related information. Moreover, an adaptive feature fusion scheme is proposed (as shown in Fig. 1), which can adaptively fuse RGB features with depth features by adding learnable weight parameters. To obtain richer local details and semantic information, we develop a detail information enhancement module and a semantic information enhancement (SIE) module. Different scale features of food can be fully utilized. Besides, we use 2960 pairs of RGB and depth images as the input to our network and perform some experiments. The experimental results demonstrate the effectiveness of our proposed method. Compared with the experimental results of RGB-D [22], our experimental results achieve the mean of 17.8%, an improvement of 0.7%. The percentage mean absolute error (PMAE) of calories, fat, carb, and protein reach 14.4%, 22.5%, 21.0%, and 19.4%, respectively, which are improved by 0.6%, 1.0%, 1.4%, and 1.6%, respectively.

The main contributions of this article are as follows.

1) We propose a multimodal and multiscale fusion model that can enhance feature information through an adaptive RGB-D fusion scheme and feature enhancement strategy.
2) We develop a detailed information enhancement (DIE) module and SIE module, which can make each level of features more semantically rich and detailed.
3) We conduct extensive experiments on the benchmark dataset Nutrition5k. The experimental results show that the proposed method can achieve promising performance, which demonstrates the efficacy of our proposed method.

The rest of this article is organized as follows. Section II introduces the related works. In Section III, the proposed method is introduced in detail. Section IV reports the experimental result. Finally, Section V concludes this article.

II. Related Works

A. Traditional Methods

Nutrition estimation is important for a healthy diet and food safety. Most existing traditional methods [29], [30], [31] of nutrition assessment are based on the inherent chemical or physical properties of nutrients in food. For example, chromatography separates and analyzes the components of a mixture by using physical properties [29], such as adsorption capacity, solubility, affinity, and retardation. In the process of chromatographic separation [32], there are two phases: mobile phase and fixed phase. According to the different mobile phases used, chromatographic methods can be divided into gas chromatography and liquid chromatography. At present, there is some research on gas chromatography [13] and liquid chromatography [14] in the field of food analysis. Through the chemical reaction between different substances, titration can be used to quantify a certain component. For example, Taprab et al. [16] used paper-based titration to analyze formaldehyde in foods. Spectrophotometry is also a commonly used method. For instance, Kalinowska et al. [7] leveraged spectrophotometry to determine the total biogenic amine content in wine. Although these methods can obtain good results, they often require additional experimental equipment or reagents and require a certain amount of specialized knowledge. In addition, most of these methods require complex preprocessing and separation of the sample to achieve the detection and analysis of some types of components in the sample. Therefore, these methods may destroy the original structure and nature of the food. Furthermore, these physicochemical methods are often used in biochemical research and industrial production, which is difficult for ordinary people to use. Therefore, these methods cannot meet the needs of simple and rapid nutrition analysis of daily foods.

B. Vision-Based Methods

In recent years, with the rapid advancements of deep learning and computer vision have given rise to numerous representative methods [33], [34], [35], [36], which are widely utilized in

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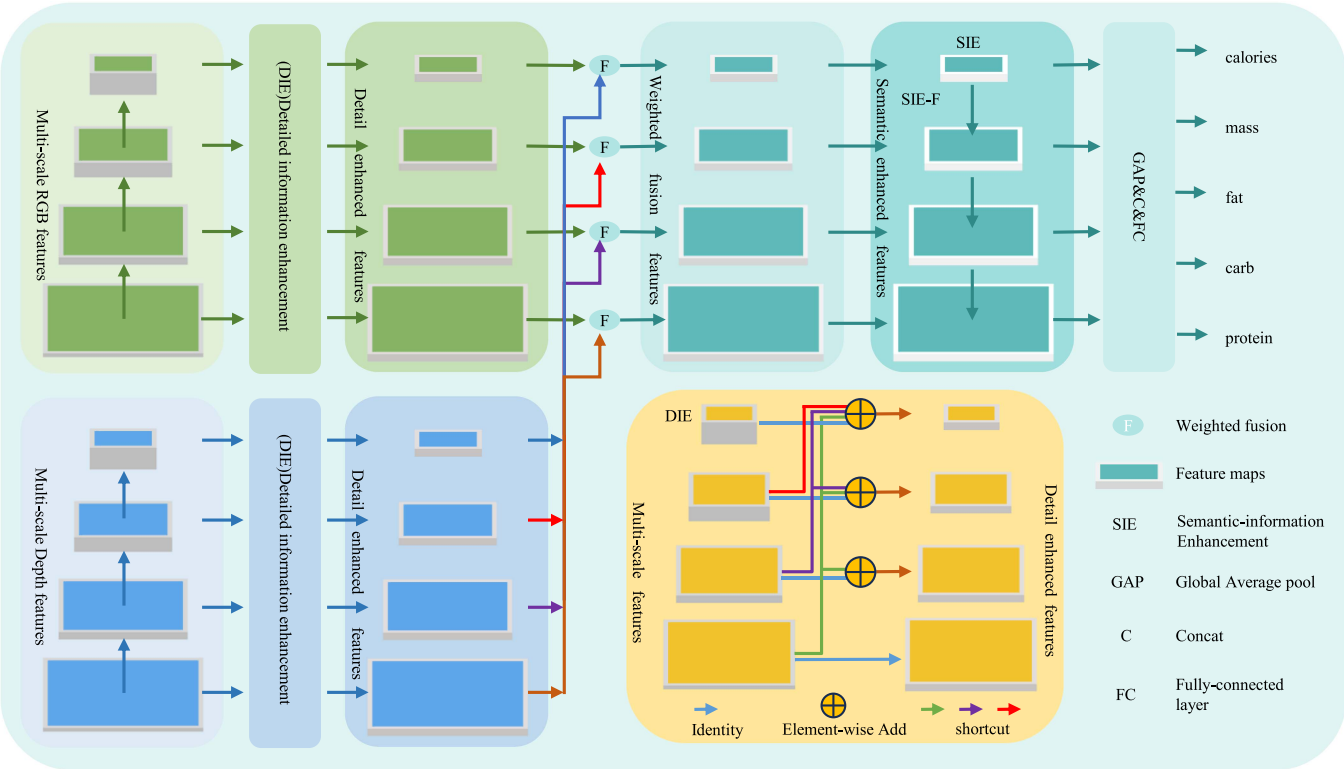

Fig. 2. Over structure of our proposed method. Our approach mainly consists of three parts: DIE, Adaptive RGB-D fusion (ADF), and SIE. "F" indicates the ADF module.

diverse domains such as feature extraction [33], [37], [38], information fusion [34], [39], object detection [40], and sentiment analysis [41]. Meanwhile, vision-based methods have also found applications in the field of nutritional assessment [42], [43], [44]. For example, Ege et al. [42] used convolutional neural networks to predict calories, categories, and ingredients from food pictures. Liu et al. [43] combined nondestructive detection techniques with convolutional neural network-based computer vision systems to detect and analyze complex food matrices. Ruede et al. [44] proposed an end-to-end nutrition estimation method based on a single image and develop a large-scale food database. For the nutritional intake of hospital patients, Lu et al. [11] take pre- and postmeal RGB-D image pairs as network inputs and employ a multitask contextual network to estimate food nutrition information. Moreover, some methods [8], [21], [22] used RGB images or depth information to estimate food nutrition. For example, Swin-nutrition [21] leverages a single RGB image as input and combines it with FPN to obtain the predicted nutrients. However, Swin-nutrition only utilizes RGB images, some spatial information may be ignored. Afterwards, Thames et al. [8] adopted the depth images as the fourth channel, which is sampled as a three-channel tensor and input to the model. However, RGB image and depth information are not fully utilized and fused. To this end, Shao et al. [22] proposed an RGB-D model, which integrates FPN, BPN, and CBAM to fuse RGB features with depth image features. Besides, Open-Vocabulary Segmentation Nutrition (OVS-Nutrition) [45] and Ingredient-Guided Nutrition (IG-Nutrition) [46] leverage auxiliary information to assist nutrition estimation. Concretely, OVS-Nutrition leverages the segment information to better delineate food, IG-Nutrition utilizes ingredient information to enable a more granular analysis of nutritional content. By incorporating auxiliary information, both methods can achieve better nutrition estimation results. Although the abovementioned methods can achieve a good performance, there are still some issues that need to be further considered. For example, most existing methods may not fully utilize different scale features and some detailed information may not be well explored, leading to suboptimal performance.

To solve the abovementioned issues, we develop an adaptive feature fusion and feature enhancement method for food nutrition estimation. In the proposed method, a weighted RGB-D fusion module is given to adaptively fuse RGB-D features. In addition, we introduce a DIE module and an SIE module, which can enhance both semantic information and detail information.

III. Method

The overall framework of our proposed method is illustrated in Fig. 2. First, for the input pairs of RGB image and depth image, we extract the image features with different scales by the pretrained image encoder. Then, original features are fed into the DIE module. In addition, we adaptively fuse the RGB features and depth image features with the weighted RGB-D fusion module. Furthermore, we design an SIE module, which can obtain more semantic information by using multiscale features. Finally, we use a multitask learning loss function to optimize the model. The implementation details of these modules are described in the following sections.

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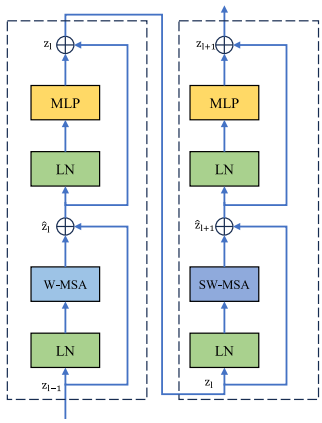

Fig. 3. Specific structure of swin-transformer block. W-MSA denotes window-based self-attention, and SW-MSA represents shifted window-based self-attention. $\hat{z}_l$ and z_l denote the output feature map of the ith block after SW-MSA and MLP, respectively. LN indicates the layer norm operation and MLP stands for multilayer perceptron.

A. Vision Encoder

Give a pair of input RGB images and a depth image, the vision encoder aims to extract feature representations. In this work, we adopt swin transformer as our backbone network, which can combine the advantages of both CNN and transformer. To be specific, the input images are first divided into nonoverlapping patches and then linearly mapped. Subsequently, these patches are fed into the swin-transformer block. The specific structure of the swin-transformer block is shown in Fig. 3. We can see that the block utilizes window multihead self-attention (W-MSA) and shifted-window multihead self-attention (SW-MSA), which can discover local and global information with less computation. Besides, swin-transformer leverages self-attention scheme, which can focus on more effective features. Moreover, the last layer of features undergoes more self-attention operations and the receptive field continues to expand, the features in the last layer can contain more global semantic information. In contrast, the front layer features have more local detailed information. Therefore, the local and global information can be well extracted. In addition, features at different scales have an impact on nutritional assessment. Thus, we leverage swin-transformer to initially extract the RGB image features I_{rgb} and depth image features I_{depth} of different levels as follows:

$$\begin{aligned} I_{\text{rgb}} &= IE_{\text{rgb}}\left(\text{Img}_{\text{rgb}}\right) \\ I_{\text{rgb}} &= \left(I^1_{\text{rgb}}, I^2_{\text{rgb}}, \ldots, I^n_{\text{rgb}}\right) \end{aligned} \tag{1}$$

where IE_{rgb} denotes the RGB image encoder, which processes the input image Img_{rgb} to extract hierarchical features. Besides, n is 4 in this work. The ith level feature I^i_{rgb}, is defined as $I^i_{\text{rgb}} \in R^{C_i \times H_i \times W_i}$, where C_i, H_i, and W_i are the number of channels, height, and width of the feature maps, respectively. The encoder follows a four-stage hierarchical structure, in which the number of channels C doubles with each successive stage, while the spatial resolution is reduced by half. Consequently, two sets of multiresolution feature maps, I_{rgb} and I_{depth}, are generated for both modalities. The I_{depth} is similar to I_{rgb}.

Although swin transformer has powerful performance in local and global feature extraction, there are smaller ingredients in the food nutrition estimation task that are difficult to detect, which may degrade the performance. Considering the abovementioned reason, it is necessary to extract local detailed information from the image. Therefore, the DIE module is proposed in this work.

B. Detail-Information Enhancement

By using the image encoder, we can obtain four-level features. Since the features at different levels focus on different information, shallow features contain more detailed information, while deep features contain more semantic information. Therefore, we build a DIE module to enhance detailed information by utilizing the shallow feature. For the different level RGB features $I_{\text{rgb}} = (I^1_{\text{rgb}}, I^2_{\text{rgb}}, I^3_{\text{rgb}}, I^4_{\text{rgb}})$, we first group them into the same channel dimension, which can be represented as follows:

$$I^i_{\text{rgb-c}} \in R^{C \times H_i \times W_i} = \text{conv}_i\left(I^i_{\text{rgb}}\right) \tag{2}$$

where $I^i_{\text{rgb-c}}$ is the ith level RGB feature, which has the same channel dimension C (C is set to 256). conv_i is 1×1 convolution kernel. Next, we leverage shallow features to enhance other level features as follows:

$$\begin{cases} I^i_{\text{rgb-d}} = I^1_{\text{rgb-c}}, & \text{if } i = 1 \\ I^i_{\text{rgb-d}} = \sum_{j=1}^{i-1} \phi_j\left(I^j_{\text{rgb-c}}\right), & \text{otherwise} \end{cases} \tag{3}$$

where $I^i_{\text{rgb-d}}$ is the ith level RGB feature, which can be enhanced by detailed information, $i \in [1, 4]$ and $j \in [1, i-1]$. The downsampling operation for the jth level feature denoted as ϕ_j, is implemented using a 3×3 convolution kernel with a stride of 2. The proposed DIE leverages the detailed information in large-scale feature maps to augment the smaller-scale feature maps, which contain rich local information, to augment deeper layers focusing on semantic understanding, making it ideal for tasks requiring precise local information.

C. Adaptive RGB-D Fusion

After processing by the DIE module, we can obtain multilevel RGB features $(I_{\text{rgb-d}} = (I^1_{\text{rgb-d}}, \ldots, I^n_{\text{rgb-d}}))$ and depth image features $(I_{\text{depth-d}} = (I^1_{\text{depth-d}}, \ldots, I^n_{\text{depth-d}}))$ are rich in detailed information. Because the RGB feature and depth image feature come from different sources and contain different information, their contribution to the final fusion features is also different. To solve this problem, we develop an adaptive RGB-D fusion module (ADF), as shown in Fig. 4. Specifically, we use an additional weight for each input feature to adaptively fuse features from different modalities. The process can be described as follows:

$$I^i_f = \frac{w_{i,1}}{w_{i,1} + w_{i,2} + \epsilon} \cdot I^i_{rgb-d} \oplus \frac{w_{i,2}}{w_{i,1} + w_{i,2} + \epsilon} \cdot I^i_{\text{depth-d}} \tag{4}$$

where $w_{i,1}, w_{i,2} \geq 0$, and I^i_f is the ith level fusion feature. $w_{i,1}, w_{i,2}$ are the weights of the RGB feature and depth image feature, respectively. ϵ is a small value to avoid numerical instability. $\oplus$ denotes element-wise add operation. Two learnable parameters are assigned as weights to the RGB and depth

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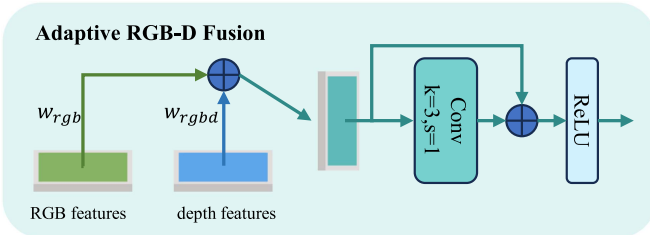

Fig. 4. Ours Adaptive RGB-D Fusion module detail. The green block is the RGB feature, the blue is the depth feature, and the cyan is the fusion feature. W_{rgb} and W_{rgbd} are the weight parameters of rgb and depth features for adaptive fusion.

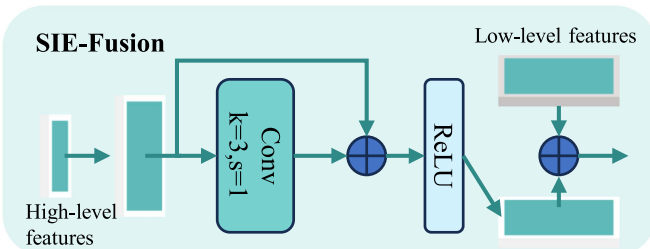

Fig. 5. Details of the fusion method for high-level features and low-level features in SIE. The small size cyan block on the left is the high-level feature, and the large size cyan block on the upper right is the low-level feature.

features, respectively, and are dynamically updated throughout the model training process.

D. Semantic-Information Enhancement

By using the Adaptive RGB-D fusion scheme, we can obtain different levels of fusion feature $I_f = (I_f^1, \ldots, I_f^n)$. High-level fusion features are rich in detailed information through the DIE module, while the low-level features lack semantic information. In order to make low-level features have rich semantic information, the SIE module, as illustrated in Fig. 5, is designed as follows:

$$I_{up}^i = \text{ReLU}\left(\theta(I_s^i) \oplus \phi_i(\theta(I_s))\right)$$
$$\begin{cases} I_s^i = I_f^i, & \text{if } i = n \\ I_s^i = I_f^i \oplus I_{\text{up}}^{i+1}, & \text{other} \end{cases} \tag{5}$$

where θ is the bilinear interpolation operation that magnifies the feature scale. ϕ_i is a convolution layer with kernel size 3 and stride 1. ReLU is the nonnegative nonlinear activation function. $\oplus$ indicates element-wise operation. n is 4 in our method. The abovementioned scheme ensures that low-level features gain semantic richness while retaining inherent detailed information. Besides, the SIE introduces a structured element-wise fusion process where each level is refined by combining interpolated features with additional convolutional refinements, which can obtain a more balanced representation across feature levels.

E. Multitask Learning

By leveraging the proposed DIE, RGB-D fusion, and SIE module, we can get multilevel features $(I_s = (I_s^1, \ldots, I_s^n))$, which contain rich details and semantic information. Besides, it is very costly to directly use the different level feature map for nutrition estimation. Therefore, we adopt global average pooling operation (GAP) and linear layer to reduce the cost and obtain the result as follows:

$$I_{re}^i = w_3 \cdot (w_2 \cdot \text{ReLU}(w_1 \cdot \text{GAP}(I_s^i) + b_1) + b_2) + b_3 \tag{6}$$

where w_1, w_2, w_3, b_1, b_2, and b_3 are the parameter of the linear layers. $i \in [1, 4]$. Finally, for the five predicted values (calories, mass, fat, carb, protein), we use a multitask loss function to optimize the model. The loss function is given as follows:

$$l_{\text{cal}} = \frac{\sum_{i=1}^{n} \left|\hat{y}_i^{\text{cal}} - y_i^{\text{cal}}\right|}{\sum_{i=1}^{n} y_i^{\text{cal}}}$$
$$L_{\text{total}} = l_{\text{cal}} + l_{\text{mass}} + l_{\text{fat}} + l_{\text{carb}} + l_{\text{protein}} \tag{7}$$

where y_i^{cal} represents the true calories value of ith food image, and $\hat{y}_i^{\text{cal}}$ denotes the predicted value. The loss function for calorie prediction is represented as l_{cal}. Similarly, the objective functions for l_{mass}, l_{fat}, l_{carb}, and l_{protein} are defined in the same way as l_{cal}. By combining the abovementioned objective functions, we can obtain the overall objective function L_{total}.

IV. Experiments

A. Dataset and Setting

1) Dataset: The Nutrition5k dataset [8] consists of more than 20k short videos and 3.5k RGB-D image pairs generated from roughly 5000 unique dishes. Each dish in this dataset contains ingredient information, mass and heat of each ingredient, and some nutrition information, such as fat, carb, and protein quality. As in previous works, i.e., RGB-D [22] and Swin-Nutrition [21], the dataset is divided into the train set and the test set by 5:1. We adopt the same experiments in our study. Some samples from nutrition5k are shown in Fig. 6.

2) Experimental Setup: In the proposed method, we utilize the swin-transformer model as our vision encoder. Swin-transformer has four different model variants, namely Swin-T, Swin-S, Swin-B, and Swin-L. Considering the complexity of the model and the size of the dataset, we select Swin-B as the backbone network of our model. And the Swin-B is available in two versions, Swin-B-224 and Swin-B-384, which are pretrained on image size 224 and 384 imageNet-1k dataset, respectively. We use both versions of Swin-B to initialize our model. When encoding the images, we adopt the image resizing, centre cropped process. By using Swin-B, we can obtain four levels of feature, the channel and size of features are (128,256,512,1024) and $(\frac{H}{4}, \frac{H}{8}, \frac{H}{16}, \frac{H}{32})$, respectively. In the proposed DIE, the channel dimension of the four levels of features is set to the same 256. Our final linear layer output dimension is 1024. We employ Adam as the optimizer and conduct all the experiments on an NVIDIA RTX 3090 GPU with 24GB of RAM. For Swin-B-224, the batch size is set to 32. For Swin-B-384, the batch size is set to 8. The number of training epochs is 150, and the initial learning rate is set to 10^{-4}. During the training, we use the cosine annealing warm restarts schedule and the learning rate is varied between 10^{-4} and 10^{-6}.

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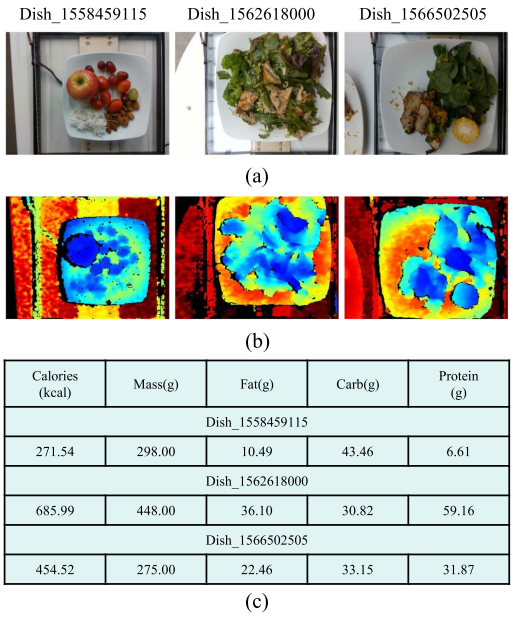

Calories (kcal)	Mass(g)	Fat(g)	Carb(g)	Protein (g)
Dish_1558459115				
271.54	298.00	10.49	43.46	6.61
Dish_1562618000				
685.99	448.00	36.10	30.82	59.16
Dish_1566502505				
454.52	275.00	22.46	33.15	31.87

(c)

Fig. 6. (a) Food RGB images. (b) Depth images associated with RGB images. (c) Nutritional content annotations associated with images.

3) Evaluation Metrics: In the nutrition estimation task, MAE and PMAE are the commonly used metrics to evaluate and predict performance. Following the previous works [8], [22], our proposed method is also tested using two metrics: MAE and PMAE, which reflect the error between the predicted results of the model and the true values. Lower MAE and PMAE values indicate better prediction. MAE and PMAE are defined as follows:

$$\text{MAE} = \frac{1}{N}\sum_{i=1}^{N}|\hat{y}_i - y_i| \tag{8}$$

$$\text{PMAE} = \frac{\text{MAE}}{\frac{1}{N}\sum_{i=1}^{N} y_i} \tag{9}$$

where N is the number of samples, y_i indicates the groundtruth of ith sample, and $\hat{y}_i$ is the predicted value of ith sample.

B. Performance Comparison

In this section, some experiments are performed to verify the efficacy of the proposed method. Since there are few models specifically designed for nutritional assessment, similar to other methods [21], [22], we also select some networks for performance comparison in nutritional assessment. The selected networks include AlexNet [47], VGG-16 [48], Inception V3 [49], and ResNet [26] networks, which are the mainstream backbone networks, and CoTNet [50], which is a variant of ResNet. The abovementioned networks adopt a two-branch network structure, and the two branches are used to extract the features of RGB and depth images, respectively. The multimodal fusion of the abovementioned networks leverages simple feature vector concatenation. For Swin-B, we also utilize a two-branch network structure, and the two branches are used to extract the features of RGB and depth images, respectively. The multimodal fusion scheme is similar to the abovementioned methods. In addition, we also select some existing competitive nutrition estimation methods, i.e., Google-Nutrition [8], RGB-D [22], Swin-Nutrtion [21], Portion-Nutrition [53], Coarse-to-Fine Nutrition [54], and RoDE [55].

Some experiments are conducted to compare our approach with other existing methods. The comparison results are shown in Table I and Table II. For our method, we use detail enhancement, semantic enhancement, and adaptive fusion of multimodal features to obtain more important features. As we can see, our method consistently outperforms other approaches in most cases, achieving PMAE values of 14.4%, 21.0%, and 19.4% for calories, carbohydrate, and protein, respectively. For Google-nutrition [8], the method uses Inception V2 as the backbone network to predict the content of five nutrients and takes RGB and depth images as input. The method predicts a caloric PMAE of 18.8%, a mass PMAE of 18.9%, and a protein PMAE of 20.9%. Our method is compared with the experimental results of Google-Nutrition [8], all values surpass it except for fat. The mean value reached 17.8%, exceeding Google-Nutrition by 2.3%. For RGB-D [22], the method uses two branches Resnet to extract RGB image features and depth image features, and fuses features through multiscale fusion and attention strategy. The method predicts a caloric PMAE of 15.0%, a mass PMAE of 10.8%, a fat PMAE of 23.5%, a carb PMAE of 22.4%, and a protein PMAE of 21.0%. Compared with the RGB-D, all values of our method are better than RGB-D except for mass. The mean value reaches 17.8% and exceeds it by 0.7%. Moreover, our method also outperforms Swin-Nutrition [21] by 1.6% in average PMAE value. Compared to portion-nutrition [53], the developed method achieves 1.4% improvement in calories PMAE value. It is worth noting that our reported values are directly obtained from the corresponding articles. In cases where certain methods lack results, it is because the original articles are not provide the relevant data. For coarse-to-fine nutrition [54], our method exceeds it by 11.2% in average PMAE value. Finally, when compared with RoDE [55], our method surpass it by 34.1% in mean PAME value. From the abovementioned results, we can see that the proposed method can achieve a good performance, which can be primarily attributed to the design of the DIE module, ADF module, and SIE module. More specifically, the DIE module enhances the details of the features at each scale, enabling the model to focus more effectively on small target objects. Besides, our ADF module can adaptively fuse RGB and depth features, leading to more discriminative features, while the proposed SIE module can enhance nutrient prediction by leveraging the SIE scheme, further improving prediction results. These combined innovations enable our method to consistently achieve good results.

C. Visualization Analysis

To visualize our prediction results more clearly, we compare the predicted value (Y-axis) from our proposed method with the groundtruth value (X-axis). The results are shown in Fig. 7. Besides, we compare our method against the competitive methods, Swin-nutrition and RGB-D. Concretely, the green line

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TABLE I
COMPARISON OF DIFFERENT METHODS ON PMAE(%) VALUES

Method	Calories PMAE (%)	Mass PMAE (%)	Fat PMAE (%)	Carb PMAE (%)	Protein PMAE (%)	Mean PMAE (%)
ALexNet [47]	19.0	18.2	30.1	29.2	29.0	25.1
VGG-16 [48]	19.2	17.9	29.6	29.0	28.4	24.8
Inception-V3 [49]	19.5	16.8	29.0	28.7	28.5	24.5
ResNet-50 [26]	19.0	16.2	28.9	28.1	28.3	24.1
ResNet-101 [26]	18.3	14.9	28.4	27.9	27.6	23.4
CoTNet [50]	18.4	15.1	29.1	28.0	27.9	23.7
ACNet [51]	18.1	17.2	30.4	28.1	27.3	24.2
RDFNet [52]	18.2	15.7	31.7	29.0	27.6	24.4
Swin-B [28]	16.4	12.9	25.8	22.0	22.5	19.9
Swin-B† [28]	16.4	12.7	25.0	21.9	22.6	19.7
Google-nutrition [8]	18.8	18.9	**18.1**	23.8	20.9	20.1
RGB-D [22]	15.0	**10.8**	23.5	22.4	21.0	18.5
Swin-nutrition [21]	16.3	12.9	24.9	21.2	21.7	19.4
Portion-nutrition [53]	15.8	-	-	-	-	-
Coarse-to-fine nutrition [54]	24.1	19.4	36.0	32.1	33.5	29.0
RoDE [55]	52.4	38.4	67.1	47.8	53.9	51.9
ADFE † (Ours)	14.9	11.7	23.8	21.7	21.4	18.7
ADFE (Ours)	**14.4**	11.7	22.5	**21.0**	**19.4**	**17.8**

The best results are bold in black. The patch size and input image size of Swin-B is 7 × 7 and 224 × 224. The patch size and input image size of Swin-B† are 12 × 12 and 384 × 384. Ours models(ADFE† and ADFE) use Swin-B and Swin-B† as backbone network respectively.

TABLE II
COMPARISON OF DIFFERENT METHODS ON MAE VALUES

Method	Calories MAE (kcal)	Mass MAE (g)	Fat MAE (g)	Carb MAE (g)	Protein MAE (g)
ALexNet [47]	48.77	36.39	3.88	5.78	5.09
VGG-16 [48]	49.28	35.79	3.81	5.74	4.98
Inception-V3 [49]	50.06	33.59	3.73	5.68	5.00
ResNet-50 [26]	48.77	32.39	3.72	5.56	4.97
ResNet-101 [26]	46.97	29.79	3.66	5.52	4.84
CoTNet [50]	47.23	30.19	3.75	5.54	4.90
ACNet [51]	46.46	34.39	3.88	5.56	4.78
RDFNet [52]	46.72	31.39	4.08	5.74	4.84
Swin-B [28]	42.10	25.79	3.32	4.36	3.95
Swin-B† [28]	42.10	25.39	3.22	4.34	3.97
Google-nutrition [8]	48.26	37.79	**2.33**	4.71	3.67
RGB-D [22]	38.50	**21.59**	3.03	4.44	3.69
Swin-nutrition [21]	41.84	25.79	3.21	4.20	3.81
Portion-nutrition [53]	40.05	-	-	-	-
Coarse-to-fine nutrition [54]	61.7	38.5	4.6	6.3	5.9
RoDE [55]	-	-	-	-	-
ADFE† (Ours)	38.25	23.39	3.06	4.30	3.76
ADFE (Ours)	**36.96**	23.39	2.90	**4.16**	**3.40**

The best results are bold in black.

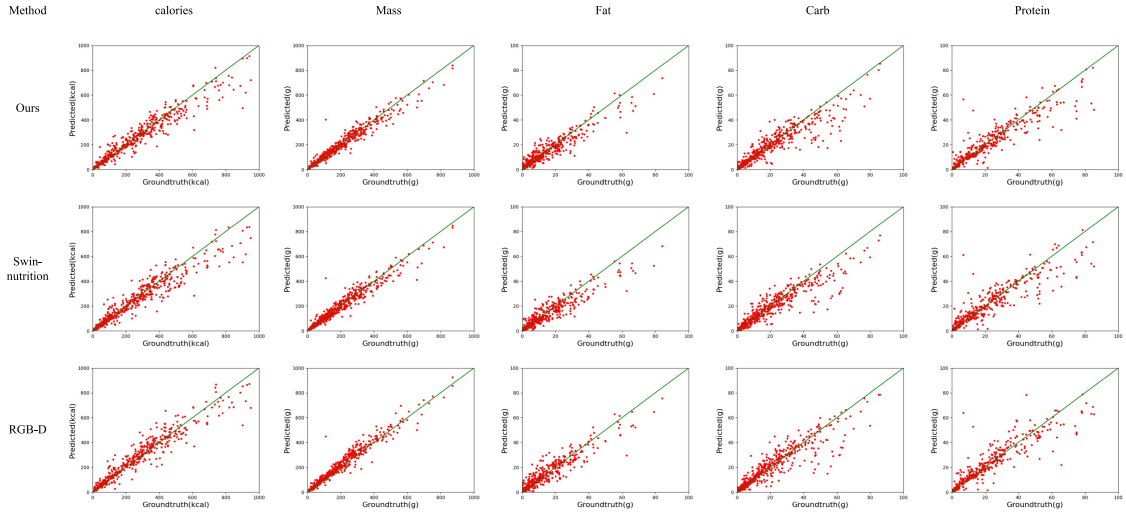

Fig. 7. Relationships between the prediction values and the groundtruth values from our proposed method and other methods.

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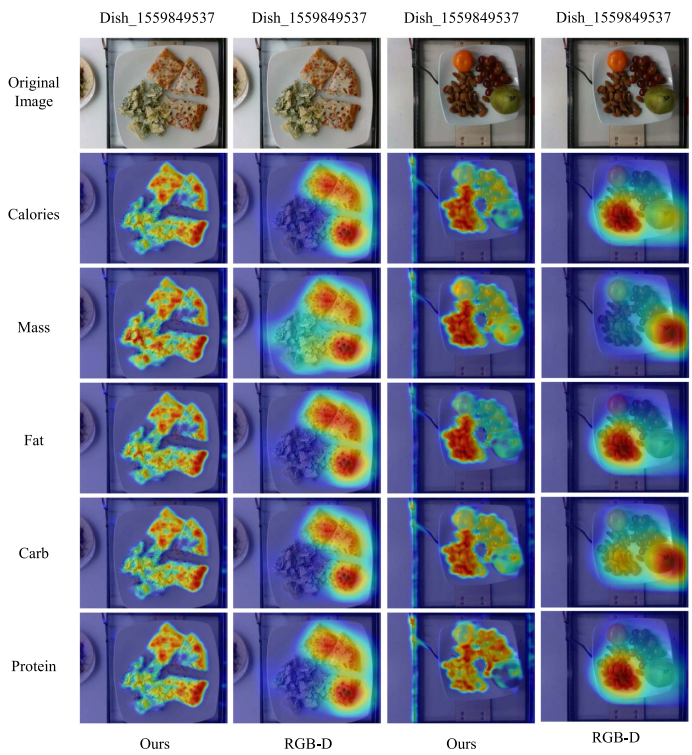

(a)

Dish_1559849537					
Ingredient	Calories (kcal)	Mass(g)	Fat(g)	Carb(g)	Protein (g)
Caesar salad	44.44	101.00	2.12	4.34	3.23
Cheese pizza	390.873	147.00	13.82	49.25	16.46
Sum	435.31	248.00	15.94	53.59	19.69
Dish_1559849537					
Ingredient	Calories (kcal)	Mass(g)	Fat(g)	Carb(g)	Protein (g)
Grapes	53.82	78.00	0.16	14.04	0.55
Pears	114.97	201.00	0.201	30.55	0.80
Mandarin Oranges	28.09	53.00	0.16	6.89	0.42
Almonds	352.64	61.00	30.20	13.85	13.66
Sum	549.52	393.00	30.71	65.33	15.44

(b)

Method	Calories (kcal)	Mass(g)	Fat(g)	Carb(g)	Protein (g)
Dish_1559849537					
RGB-D	454.9	240.7	16.6	57.2	19.7
Ours	443.7	229.8	16.4	54.3	19.9
Truth	435.31	248.00	15.94	53.59	19.69
Dish_1559849537					
RGB-D	599.5	441.6	30.9	68.5	18.2
Ours	583.3	428.7	33.5	64.6	17.3
Sum	549.52	393.00	30.71	65.33	15.44

(c)

Fig. 8. (a) Visualization of the prediction results for our method and RGB-D using Grad-CAM. (b) Nutritional content annotations associated with the original image. (c) Prediction results.

indicates that the predicted value is the same as the true value, the closer the scatter points are to the green line, the more accurate the results. We can see that for calories, fat carbs and protein, the predicted values of our method are more compact and closer to the green line than that of swin-nutrition and RGB-D. We can see our proposed method and existing methods appear minimal. However, Swin-nutrition or RGB-D perform not well in some metrics. For example, Swin-nutrition shows slightly larger deviations at higher protein and fat values, where its predictions tend to underperform. Similarly, RGB-D shows a broader scatter in calorie and carb predictions, particularly at mid-to-high values.

In addition, we utilize the grad-cam to visualize the regions of interest of the model and compare them with RGB-D. The experimental results are shown in Fig. 8. We can see that our method exhibits more detailed attention in the region of interest compared to RGB-D. For example, in Dish1559849537, our approach focuses on more precise areas that closely correspond to the regions where nutrients are located. In the calories heat map, the dish table shows that Caesar salad, which contains fewer calories and receives less attention. These observations underscore the strengths of our method.

D. Ablation Study

In this section, we perform some ablation experiments to explore the influence of different modules. The baseline of our approach is a dual pretrained swin-B, and the four levels of RGB features are fused by directly adding the depth features.

First, we analyze the effects of pretrained models on the imageNet1k dataset using images of different sizes on nutritional assessment. The experimental results are given in Table III, we can see the mean PMAE value of Swin-B backbone network (Base-1) is lower than that of Swin-B† network (Base-2). Compared with the Swin-B, the Swin-B† shows improvements in PMAE for calories, fat, carbohydrates, protein, and the mean by 0.7%, 1.3%, 0.3%, with reductions of 0.6%, and 0.4%, respectively. In addition, after adding other modules to the baseline, the average PMAE values of DIE-2, DA-2, and adaptive feature fusion and enhancement (ADFE) are also higher than DIE-1, DA-1, and ADFE†, respectively. This shows that the size of the input image and patch size can affect the final result. Although large-size images provide more information, they also increase computational costs, and the performance gains diminish beyond a certain size threshold. Second, we analyze the impact of each module. The DIE module is a multiscale detail enhancement module that enriches feature representation across different levels by incorporating high-resolution features into low-resolution features, ensuring detailed representations across scales. In Table III, after adding the DIE module on Base-1 and Base-2, the mean PMAE of DIE-1 is reduced by 0.6% compared to Base-1, the mean PMAE of DIE-2 is similarly 0.6% lower than Base-2. The experimental results show the positive contribution of the DIE module to performance enhancement. Besides, by adding the ADF module to DIE-1 and DIE-2. We can see that the mean PMAE values of DA-1 and DA-2 reach 19.1% and 18.3%, respectively. The

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TABLE III
PERFORMANCE COMPARISON RESULTS FOR EACH MODULE ON THE NUTRITION5K DATASET

Method	Module			backbone		Nutrition-5k					
	DIE	ADF	SIE	Swin-B	Swin-B†	Calories PMAE (%)	Mass PMAE (%)	Fat PMAE (%)	Carb PMAE (%)	Protein PMAE (%)	Mean PMAE (%)
Base-1				✓		16.4	12.9	25.8	22.0	22.5	19.9
DIE-1	✓			✓		15.8	12.1	24.5	22.1	21.9	19.3
DA-1	✓	✓		✓		15.9	11.7	24.7	21.4	21.7	19.1
ADFE†	✓	✓	✓	✓		14.9	11.7	23.8	21.7	21.4	18.7
Base-2					✓	16.4	12.7	25.0	21.9	22.6	19.7
DIE-2	✓				✓	15.8	12.7	23.9	22.0	21.0	19.1
DA-2	✓	✓			✓	14.7	**11.6**	23.4	22.0	20.0	18.3
ADFE	✓	✓	✓		✓	**14.4**	11.7	**22.5**	**21.0**	**19.4**	**17.8**

The best results are shown in boldface. 'DIE' is the detailed-information enhancement, 'ADF' denotes the adaptive RGB-D fusion, 'SIE' represents the semantic information enhancement. Swin-B and Swin-B† are different with the input image size and patch size.

TABLE IV
PERFORMANCE COMPARISON OF EACH MODULE WITH SWIN-B AS BACKBONE NETWORK

Method	Module			Nutrition-5k					
	DIE	ADF	SIE	Calories PMAE (%)	Mass PMAE (%)	Fat PMAE (%)	Carb PMAE (%)	Protein PMAE (%)	Mean PMAE (%)
Base-1				16.4	12.9	25.8	22.0	22.5	19.9
DIE-1	✓			15.8	12.1	24.5	22.1	21.9	19.3
ADF-1		✓		15.3	12.0	23.2	21.3	22.0	18.8
SIE-1			✓	15.4	11.8	24.6	21.1	21.2	18.8
DA-1	✓	✓		15.9	11.7	24.7	21.4	21.7	19.1
AS-1		✓	✓	15.9	11.5	24.1	21.0	22.0	18.9
DS-1	✓		✓	15.3	11.3	24.4	22.5	20.7	18.8
ADFE†	✓	✓	✓	14.9	11.7	23.8	21.7	21.4	18.7

TABLE V
OUR ADAPTIVE RGB-D FUSION METHOD COMPARES TO THE BASIC FUSION METHOD

Method	Calories PMAE (%)	Mass PMAE (%)	Fat PMAE (%)	Carb PMAE (%)	Protein PMAE (%)	Mean PMAE (%)
Add	17.0	12.7	26.0	22.4	23.6	20.3
Concat	16.4	12.5	25.2	23.5	22.0	19.9
Ours (ADF)	**15.3**	**12.0**	**23.2**	**21.3**	**22.0**	**18.8**

All use Swin-B as the vision encoder.
The best results are marked in bold.

TABLE VI
USE RESNET AS OUR VISION ENCODER COMPARED WITH OUR ORIGINAL VISION ENCODER (SWIN-B†)

Method	Calories PMAE (%)	Mass PMAE (%)	Fat PMAE (%)	Carb PMAE (%)	Protein PMAE (%)	Mean PMAE (%)
ResNet-50	19.0	16.2	28.9	28.1	28.3	24.1
ADFE (ResNet-50)	16.4	12.6	24.7	24.5	21.8	20.0
ResNet-101	18.3	14.9	28.4	27.9	27.6	23.4
ADFE (ResNet-101)	15.1	10.3	22.6	21.1	20.5	17.9
Swin-B†	16.4	12.7	25.0	21.9	22.6	19.7
ADFE (Swin-B†)	14.4	11.7	22.5	21.0	19.4	17.8

TABLE VII
COMPARISON OF THE PROPOSED METHOD WITH OTHERS IN TERMS OF PARAMETERS(PARAMS), COMPLEXITY(FLOPS)

Method	Params (M)	FLOPs (G)
ACNet	129.61	72.10
RDFNet	140.56	78.83
RGB-D	110.87	68.96
Ours (SWin-B†)	201.89	43.15
Ours (SWin-B)	210.89	126.94
Ours (ResNet-101)	114.52	84.98

abovementioned results show that adaptive learning of the relationships between RGB features and depth features can obtain more effective fusion features. The SIE module is an FPN-like structure that adds high-level semantic information to shallow features through a top-down process. The integration of the SIE module and DIE module ensures the balance between detailed and semantic information. By incorporating all three modules, the mean PMAE value reaches the best performance, highlighting the positive impact of these components. The abovementioned results further validate the effectiveness of our proposed approach.

In addition, we examine the effect of different module combinations on the results. As shown in Table IV, the DIE, ADF, and SIE modules are individually added to the baseline, yielding mean PMAE values of 19.3%, 18.8%, and 18.8% for DIE-1, ADF-1, and SIE-1, respectively. From the results, we can see that semantic information is more important, and the adaptive fusion of RGB features and depth image features also plays a positive impact on performance. Besides, compared to DIE-1,

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the mean PMAE value of DA-1 decreases by 0.2% × 0.2%. The performance of DA-1 is slightly worse than that of ADF-1. This may be due to excessive detail enhancement leading to some global semantic information loss. Furthermore, compared to AS-1, the mean PMAE value of ADFE† decreases by 0.2%, indicating that the DIE module also contributes positively to nutrition estimation. Therefore, these results demonstrate that each of the three modules has a positive effect on enhancing the accuracy of nutritional evaluation.

To prove the effectiveness of the multimodal fusion scheme, we compare it with some basic multimodal fusion schemes (the add and concatenation schemes), and the results are shown in Table V. The results demonstrate that our method achieves the best results, which validates the effectiveness of our multimodal fusion method (ADF). Moreover, in Table VI, we use ResNet-101 as the vision encoder, achieving the mean PMAE value of only 0.1% lower than that obtained with Swin-B as the vision encoder. Applying the proposed method to ResNet-50 and ResNet-101 improves the average PMAE value by 4%–5%. Besides, compared with Swin-B† (the experimental settings are similar to [22]), we can see the performance of our method is better than Swin-B†, which validates the efficacy of the developed ADFE.

E. Computational Efficiency

In this section, we compare the proposed ADFE with some methods in terms of the parameters (Params) and complexity (FLOPs). The results are shown in Table VII. We can see that our method (ResNet-101) ranks second among the compared methods in terms of the parameters and FLOPs. The number of parameters and FLOPs of our method are 210.89 M and 126.94 G, respectively. The reason for the large number of parameters of our method (Swin-B, Swin-B†) is primarily due to the swin-transformer visual encoders, which contribute 173.36 M parameters. Despite this, our method achieves promising predictive accuracy while maintaining comparable efficiency, demonstrating good overall performance.

V. Conclusion

In this article, we propose a novel vision-based approach for nutrition estimation, named ADFE. The proposed ADFE combines RGB images and depth images to learn comprehensive feature information about evaluated food. Besides, the detailed-information enhancement, weighted RGB-D fusion, and SIE scheme are developed in this method, which can generate more discriminative multilevel features and achieve effective feature fusion. The proposed model is further optimized using a multitask learning loss. Comprehensive experiments on the Nutrition5K demonstrate that our method achieves accurate nutritional assessment, thereby validating the effectiveness of the proposed method. Although our ADFE has shown promising performance, the proposed model may not yet be suitable for direct deployment on mobile devices. In the future, we plan to develop a more lightweight model specifically designed for mobile deployment.

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Donglin Zhang received the Ph.D. degree in control science and engineering from Jiangnan University, Wuxi, China, in 2022.

He is currently an Associate Professor with the School of AI&CS, Jiangnan University. He has authored or coauthored 20 scientific articles in refereed journals and conferences, including IEEE Transactions on Knowledge and Data Engineering, IEEE Transactions on Circuits and Systems for Video Technology, IEEE Transactions on Cybernetics, ACM TOMM, IEEE Transactions on Systems, Man, and Cybernetics: Systems, and *Pattern Recognition*. His research interests include multimedia intelligence, information retrieval, and machine learning.

Boyuan Ma received the B.S. degree in computer science and technology in 2023 from Jiangnan University, Wuxi, China, where he is currently working toward the master's degree under the supervision of Mr. Donglin Zhang, with the School of AI& CS.

His research interests include food recognition, recipe retrieval, and nutrition estimation.

Xiao-Jun Wu received the B.Sc. degree in mathematics from Nanjing Normal University, Nanjing, China, in 1991 and the M.S. and Ph.D. degrees in pattern recognition and intelligent systems from the Nanjing University of Science and Technology, Nanjing, China, in 1996 and 2002, respectively.

From 1996 to 2006, he faculty member with the School of Electronics and Information, Jiangsu University of Science and Technology, where he was promoted to Professor. Since 2006, he has been with the School of Artificial Intelligence and Computer Science, Jiangnan University, Wuxi, China, where he is a Professor of Computer Science and Technology. He was a Visiting Researcher with the Centre for Vision, Speech, and Signal Processing (CVSSP), University of Surrey, Guildford, U.K., from 2003 to 2004. He has authored or coauthored more than 300 research papers in refereed international journals and conferences. His current research interests include pattern recognition, computer vision, and computational intelligence.

Dr. Wu was a fellow of the International Institute for Software Technology, United Nations University, from 1999 to 2000. He was a recipient of the Most Outstanding Postgraduate Award from the Nanjing University of Science and Technology. He is currently an Associate Editor for *Pattern Recognition Letters*, *International Journal of Computer Mathematics*, Review Editor of *Frontiers in Neurorobotics*, and Editor of *Journal of Algorithm and Computational Technology* respectively. He is a Fellow of International Association for Pattern Recognition, Asia-Pacific Artificial Intelligence Association, and Australian Institute of International Affairs.

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